IMPLEMENTATION OF FASTSPEECH 2-BASED TEXT-TO-SPEECH SYNTHESIS

Separate English and Telugu Speech Synthesis Systems

Document	Technical Project Report
Model	FastSpeech 2 + HiFi-GAN vocoder
Languages	English and Telugu
English dataset	Svarah
Telugu dataset	IndicTTS Telugu
Assignment deadline	07 November 2026

Evidence and integrity note

This report distinguishes verified outputs from incomplete or unavailable training evidence. It does not claim that the required training durations were completed where the available logs do not establish this.

Prepared as an implementation and evaluation record

Contents

1. Abstract and Objectives
2. Introduction
3. FastSpeech 2 System Architecture
4. Dataset Description
5. Preprocessing and Language-Specific Handling
6. Experimental Configuration and Training
7. Validation Results
8. Inference Test Set and Generated Outputs
9. Variance Adaptor Analysis
10. Comparative Discussion
11. Limitations and Future Work
12. Reproducibility and Project Organization
13. Conclusion

References

Appendix A. Test Sentences

> **Evidence convention**
> Values and file counts are taken from recorded project results. Missing metrics and unavailable plots are identified rather than estimated.

1. Abstract and Objectives

This project implements separate FastSpeech 2 text-to-speech inference pipelines for English and Telugu. Each system uses language-specific data preparation, text representation, configuration and checkpoint. FastSpeech 2 predicts mel spectrograms while its variance adaptor models duration, pitch and energy. HiFi-GAN converts the predicted mel representation into a waveform.

The completed inference runs produced five test audio files per language (ten WAV files in total). For each test item, duration, pitch and energy visualizations were generated, resulting in thirty variance-adaptor plots. Telugu text normalization was adjusted so that all five test sentences could be represented using the existing 66-symbol vocabulary.

Objectives

- Implement separate English and Telugu systems.
- Prepare Svarah and IndicTTS Telugu data.
- Generate five test utterances per language.
- Visualize duration, pitch and energy predictions.
- Record validation results, limitations and reproducibility details.

2. Introduction

Text-to-speech synthesis converts written language into a spoken waveform. A neural TTS system typically includes text representation, an acoustic model and a vocoder. This project uses FastSpeech 2 as the acoustic model and HiFi-GAN as the waveform generator.

FastSpeech 2 is non-autoregressive: it predicts acoustic frames in parallel after estimating how text units expand over time. Its variance adaptor introduces duration, pitch and energy information. These predictions are important for temporal alignment and prosody.

3. FastSpeech 2 System Architecture

Figure 1 summarizes the synthesis path. The variance adaptor predicts duration, pitch and energy; the length regulator expands encoded text representations to the frame-level sequence consumed by the decoder.

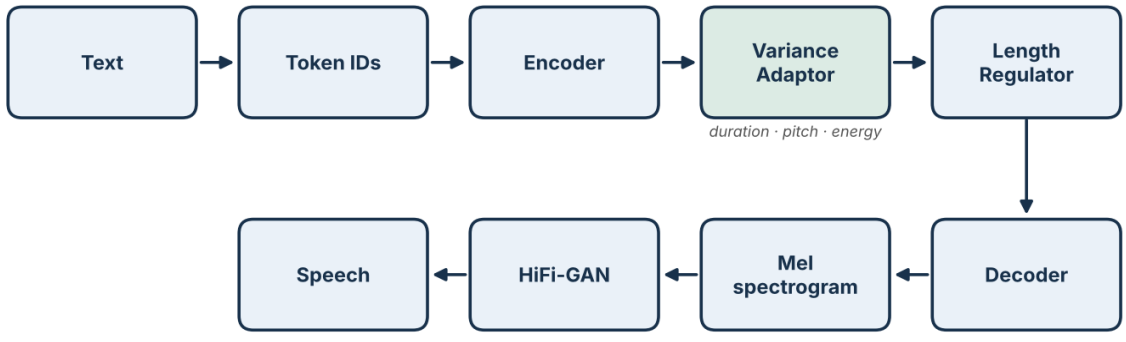


Figure 1. Conceptual FastSpeech 2 and HiFi-GAN inference pipeline.

Component	Function
Text representation	Converts the sentence into supported token or grapheme IDs.
Encoder	Produces contextual hidden representations.
Variance adaptor	Predicts duration, pitch and energy.
Length regulator	Expands encoded representations according to duration.
Decoder	Produces a mel-spectrogram representation.
HiFi-GAN	Converts mel spectrograms into waveform audio.

A duration plot helps inspect temporal allocation; pitch represents fundamental-frequency behavior and intonation; energy represents signal strength. These plots are diagnostic and do not alone prove natural or intelligible speech.

4. Dataset Description

English — Svarah: The available data contained 6,656 records and approximately 9.61 hours of audio. A custom training/validation split was prepared from the available published test partition. This is not an official Svarah train/validation split.

Telugu — IndicTTS Telugu: The prepared data contains 1,368 feature records: 1,167 training and 201 validation samples. Source audio duration was approximately five hours for training and one hour for validation. Dataset duration does not establish model-training or validation wall-clock duration.

Dataset fact	English (Svarah)	Telugu (IndicTTS)
Prepared records	6,656 source records	1,368 feature records
Duration	~9.61 hours available	~5 h train + ~1 h validation
Split	Custom split from available published test partition	1,167 train / 201 validation
Text representation	English text pipeline	66-symbol grapheme vocabulary

5. Preprocessing and Language-Specific Handling

Telugu audio was resampled to 16 kHz. Mel features used 80 channels, FFT size 1024, hop length 256, window length 1024, minimum frequency 0 Hz and maximum frequency 8 kHz. Pitch was estimated using PyWorld DIO followed by StoneMask. Energy was derived from spectrogram features.

Telugu duration labels were generated using CTC-based alignment information and a grapheme-duration heuristic. They were not manually verified phoneme alignments. This matters because duration supervision affects the length regulator and the alignment between text units and acoustic frames.

The Telugu vocabulary contains 66 symbols. Unsupported punctuation in the test sentences was normalized before inference. The date and time expression in T4 was expanded into Telugu words. All five

normalized inputs had zero unsupported characters. The original assignment sentences are retained in Appendix A; normalization applies only to model input.

The English pipeline used its separate text processing, metadata and feature paths. The precise English preprocessing parameter values should be taken from the saved English configuration when producing the code appendix; no additional values are inferred here.

6. Experimental Configuration and Training

Configuration item	English	Telugu
Model	FastSpeech 2	FastSpeech 2
Inference checkpoint	100.pth.tar (step 100)	50.pth.tar (step 50)
Vocoder	HiFi-GAN	HiFi-GAN
Text input	English text pipeline	66-symbol Telugu grapheme vocabulary
Inference	CPU-compatible	CPU-compatible

Separate language configurations and checkpoints were maintained. CPU compatibility adjustments were applied during inference and checkpoint loading to keep tensors and attention masks on compatible devices.

> **Required-duration qualification**
> The assignment specifies approximately five hours of training and one hour of validation per language. Available records do not establish completion of those full model-training and validation wall-clock durations. Telugu validation over all 201 samples took approximately 8.3 minutes. Telugu training event logs contain recorded points through step 90. English training-loss logs were unavailable.

Dataset audio duration, training wall-clock duration, validation wall-clock duration and optimizer-step count are different measurements and must not be substituted for one another.

Training curves must be based on actual training event logs. Telugu logs contain sparse recorded points through step 90, not a complete five-hour history. English training-loss logs were unavailable. English validation plots exist, but exact metric values are not reproduced here because they were not independently supplied for this report.

7. Validation Results

The recorded Telugu validation pass evaluated all 201 validation samples using checkpoint step 50 and took approximately 8.3 minutes. The aggregate losses are listed below.

Telugu validation metric	Recorded value
Total loss	41812.8009
Mel loss	5.3236
Mel postnet loss	4.5816
Pitch loss	40476.0882
Energy loss	1326.2533
Duration loss	0.5543

The pitch and energy losses are substantially larger than the mel and duration losses in the recorded aggregate. This indicates that these variance-related components require further investigation. The total loss is dominated by the large pitch and energy terms and should not be interpreted without examining individual components.

English validation results were saved in the project JSON output and validation plots were generated. Exact numeric English metrics are not included here because they were not available in the evidence used to prepare this report. Add them from the saved English validation JSON before claiming a numerical language comparison.

8. Inference Test Set and Generated Outputs

Five test items were synthesized for each language. The English set contains E1, E2a, E2b, E3 and E4. The Telugu set contains T1, T2a, T2b, T3 and T4.

Language	Test IDs	WAVs	Duration plots	Pitch plots	Energy plots
English	E1, E2a, E2b, E3, E4	5	5	5	5
Telugu	T1, T2a, T2b, T3, T4	5	5	5	5
Total	10 utterances	10	10	10	10

English audio: `output/English_test_results/audio/`. English plots: `output/English_test_results/variance_plots/`. Telugu audio: `output/Telugu_test_results/audio/`. Telugu plots: `output/Telugu_test_results/variance_plots/`. The final Telugu verification confirmed five WAV files and five plots of each feature type.

9. Variance Adaptor Analysis

Plot type	What to inspect	Caution
Duration	Relative length assigned to input units; unusually short or long regions.	Does not independently verify alignment.
Pitch	Voiced-region contour and variation.	Unvoiced regions and pitch-estimation errors can affect plots.
Energy	Relative energy over the utterance.	Scale depends on normalization and feature extraction.

Thirty plots were generated: fifteen per language. For a final visual appendix, group each utterance's duration, pitch and energy plots together, label each with language and test ID, and add a short observation based on the actual plot. The current report records plot counts but does not claim detailed visual findings from plots that were not embedded and inspected.

Audio quality should be evaluated separately by listening to each WAV. Record intelligibility, pronunciation, pauses, duration, pitch stability and audible artifacts. Do not describe speech as natural or accurate unless it has actually been assessed.

10. Comparative Discussion

Aspect	English	Telugu
Dataset	Svarah; custom split from available published test partition	IndicTTS; 1,167 train / 201 validation
Text handling	English text pipeline	66-symbol grapheme vocabulary; punctuation and date/time normalization
Checkpoint	Step 100	Step 50
Inference outputs	5 WAV + 15 plots	5 WAV + 15 plots
Validation evidence	JSON and plots; numeric values not stated here	201 samples; 8.3 min; metrics listed
Evidence limitation	Training logs unavailable	Sparse training log; high pitch/energy losses

Both systems completed the required inference output count. Their training evidence is not equally complete, and checkpoints are at different recorded steps. File counts alone are not a fair comparison of synthesis quality. A meaningful comparison requires complete metrics, consistent evaluation conditions and listening tests.

11. Limitations and Future Work

- Available logs do not verify the full five-hour training and one-hour validation wall-clock durations for both languages.

- English training-loss logs are unavailable; a verified English training curve cannot be reconstructed.
- Telugu training event logs contain sparse points through step 90.
- Telugu validation reports high pitch and energy losses, indicating weak variance prediction.
- Telugu duration labels use CTC-derived alignment and a grapheme-duration heuristic, not manually verified phoneme boundaries.
- The English split is custom-created from an available published test partition and is not an official benchmark split.
- A formal listening-based quality assessment has not been documented in the available results.

Recommended improvements

- Continue training each language independently and retain complete wall-clock, optimizer-step and validation logs.
- Save TensorBoard event files and console logs for every training run.
- Verify Telugu alignments and inspect grapheme-to-frame duration assignments.
- Investigate pitch and energy target scales, normalization, voiced/unvoiced handling and loss magnitudes.
- Record exact English validation metrics and include them in the final comparison.
- Perform a structured listening evaluation for all ten WAV files.
- Include the actual loss curves and all thirty variance-adaptor plots in the final report appendix.

12. Reproducibility and Project Organization

The main project directory is `FastSpeech2_Project_v3`. Telugu recovery assets, including prepared features, vocabulary, logs and checkpoints, are stored separately under `FastSpeech2_Telugu_Recovery`.

Artifact / directory	Purpose
train.py / preprocess.py / dataset.py	Training, preprocessing and dataset loading
synthesize.py / evaluate.py	Inference and evaluation utilities
config/Svarah	English model and preprocessing configuration
config/Telugu	Telugu model and preprocessing configuration
preprocessed_data/Svarah	Prepared English features and metadata
Telugu recovery processed/Telugu	Prepared Telugu features and metadata
output/English_test_results	English WAVs and variance plots
output/Telugu_test_results	Telugu WAVs and variance plots
output/English_validation	English validation results and plots
Telugu recovery validation_results/results	Telugu validation metrics and plots
100.pth.tar / 50.pth.tar	English / Telugu inference checkpoints

Reproducibility checklist

- Retain the exact language-specific configuration files used for each checkpoint.
- Retain matching vocabularies and speaker mappings.
- Record Python, PyTorch and dependency versions in requirements.txt.
- Record the exact checkpoint path and step for every inference run.
- Retain validation JSON files, training logs, generated audio and plots.
- Store large checkpoints and vocoder weights using Git LFS or external storage.

Suggested execution order: install dependencies → prepare datasets → preprocess acoustic features → select language configuration → train or load the corresponding checkpoint → synthesize test sentences → inspect audio and plots → save metrics and report artifacts.

13. Conclusion

Separate FastSpeech 2 inference pipelines were assembled for English and Telugu. All ten requested test WAV files and thirty variance-adaptor plots were generated. Telugu tokenization compatibility was resolved through input normalization without changing the existing 66-symbol checkpoint vocabulary.

The available evidence also identifies important work remaining. The logs do not verify the full training and validation durations specified by the assignment. Telugu pitch and energy losses are high, and duration labels are estimated. A complete final evaluation requires continued training evidence, complete loss plots, verified alignments and documented listening tests.

References

[1] Y. Ren, C. Hu, X. Tan, T. Qin, S. Zhao, Z. Zhao and T.-Y. Liu, “FastSpeech 2: Fast and High-Quality End-to-End Text to Speech,” International Conference on Learning Representations (ICLR), 2021.

[2] J. Kong, J. Kim and J. Bae, “HiFi-GAN: Generative Adversarial Networks for Efficient and High Fidelity Speech Synthesis,” Advances in Neural Information Processing Systems (NeurIPS), 2020.

[3] Svarah English speech dataset. Add the exact dataset URL, version and access date used in this project.

[4] IndicTTS Telugu speech dataset. Add the exact dataset URL, version and access date used in this project.

[5] FastSpeech 2 reference implementation. Add the exact repository URL, commit or release identifier and access date used in this project.

> **Open items**
> References [3]–[5] require the exact sources used by the project. They are left explicit rather than inventing URLs.

Appendix A. Test Sentences

Sentences are reproduced from the assignment text. Telugu text is the original assignment text; the T4 date and time expression was normalized into Telugu words only for model input (Section 5).

English

ID	Test sentence
E1	The antidisestablishmentarianism movement gained considerable momentum during the parliamentary proceedings.
E2a	The orchestra's conduct during the ceremony was praised by the audience.
E2b	The researchers will conduct an experiment to evaluate the proposed method.
E3	The rhythm of the orchestra changed unexpectedly.
E4	The meeting will start at 10:45 AM on 12 November 2026, so andaroo should arrive on time.

Telugu (original assignment text)

ID	Test sentence
T1	న్యూరోఫార్మకోలాజికల్ పరిశోధనలో నూతన ఔషధాల ప్రభావాన్ని విశ్లేషించారు.
T2a	తాడుపై అతని పట్టు చాలా బలంగా ఉంది.
T2b	ఆమె పట్టు చీరను ధరించి వేడుకకు వచ్చింది.
T3	పులి అడవిలో నిశ్శబ్దంగా నడిచింది.
T4	ఈ సమావేశం 2026 నవంబర్ 12వ తేదీ ఉదయం 10:45 గంటలకు ప్రారంభమవుతుంది, సమయానికి సిద్ధంగా ఉంచండి.

For T4 inference, the date and time were normalized into Telugu words to fit the existing vocabulary. The original sentence is retained above for traceability.